

# **KILLARA HIGH SCHOOL**



Year 12 Term 2 Assessment 2009

## Extension 2 Mathematics

Time Allowed: 70 minutes

Total marks: 45

### **Instructions**

1. Start each question in a new booklet.
2. Write on one side of the paper only.
3. Show all necessary working.

### **Question 1**

Evaluate:

a.  $\int \frac{e^{\tan x}}{\cos^2 x} dx$  1 mark

b.  $\int \frac{dx}{4 + 3\cos 2x}$  4 marks

c.  $\int \frac{3x+1}{(x-1)^2(2x+1)} dx$  3 marks

### **Question 2**

a. Evaluate  $\int_1^2 \frac{dx}{x^2\sqrt{x-1}}$  3 marks

b. Evaluate  $\int_0^1 \sqrt{\frac{3-x}{4+x}} dx$  4 marks

c. i) Use the substitution  $u = -x$  to show that  
$$\int_{-a}^0 f(x) dx = \int_0^a f(-x) dx$$
 2 marks

ii) Deduce that  $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$  2 marks

iii) Hence evaluate  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^5 \cos x dx$  2 marks

### Question 3

- a. Obtain a reduction formula for  $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$  ( $n \geq 2$ ). 3 marks
- b. Use the result from part 'a' to evaluate  $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$  3 marks

### Question 4

- a. The base of a solid is an ellipse whose major and minor axes are 16 cm and 12 cm respectively. Vertical cross sections taken through this solid in a direction parallel to the minor axis is an equilateral triangle with one side in the base of the solid. Find the volume of the solid. 5 marks
- b. The region bounded by the curve  $y = (x-2)^2$ , the x and y axes is rotated about the line  $y = -\frac{3}{2}$ . Calculate the exact volume of the solid of revolution by taking slices parallel to the y-axis. 5 marks
- c. Evaluate the exact volume obtained when the area bounded by  $y = \frac{2}{x+2}$ , the x-axis, the y-axis and the line  $x = 3$  is rotated about the y-axis. 4 marks
- d. i) Sketch the graph of  $y = e^{-t^2}$  1 mark
- ii) The region bounded by the graph the t-axis, in the domain  $-a \leq t \leq a$ , that is rotated about the y-axis forms a solid. Use the method of cylindrical shells, where a shell has radius 't' to show that the shell has an approximate volume of " $\delta V = 2\pi t e^{-t^2} \delta t$ ". 1 marks
- iii) Calculate the volume of the solid. 2 marks

## Question 1

$$a. \int \frac{e^{\tan x}}{\cos^2 x} dx$$

$$= \int \sec^2 x e^{\tan x} dx$$

$$= e^{\tan x} + C$$

✓

$$b. \int \frac{dx}{4 + 3 \cos 2x}$$

$$= \int \frac{\frac{dt}{1+t^2}}{4 + 3 \left( \frac{1-t^2}{1+t^2} \right)}$$

$$= \int \frac{dt}{4(1+t^2) + 3(1-t^2)}$$

$$= \int \frac{dt}{t^2 + 7}$$

$$= \frac{1}{\sqrt{7}} \tan^{-1} \frac{t}{\sqrt{7}} + C$$

$$= \frac{1}{\sqrt{7}} \tan^{-1} \left( \frac{\tan x}{\sqrt{7}} \right) + C$$

$$\text{let } t = \tan x \\ \text{where } \cos 2x = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} dt &= \sec^2 x dx \\ &= 1 + \tan^2 x dx \\ &= (1+t^2) dx \end{aligned}$$

$$\frac{dt}{1+t^2} = dx$$

## Question 1

C.

$$\int \frac{3x+1}{(x-1)^2(2x+1)} dx \quad \text{let } \frac{3x+1}{(x-1)^2(2x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{2x+1}$$

$$\therefore 3x+1 = A(x-1)(2x+1) + B(2x+1) + C(x-1)^2$$

$$\checkmark \text{ let } x=1 \Rightarrow B = \frac{4}{3}$$

$$\text{let } x = -\frac{1}{2} \Rightarrow C = -\frac{2}{9}$$

Equating coefficients of  $x^2$

$$A = \frac{1}{9}$$

$$\therefore \int \frac{3x+1}{(x-1)^2(2x+1)} dx$$

$$= \int \frac{1}{9(x-1)} dx + \frac{4}{3} \int \frac{dx}{(x-1)^2} - \frac{2}{9} \int \frac{dx}{2x+1} \quad \checkmark$$

$$= \frac{1}{9} \ln(x-1) + \frac{4}{3} \left( \frac{-1}{x-1} \right) - \frac{2}{18} \ln(2x+1) + C$$

$$= \frac{1}{9} \ln \left| \frac{x-1}{2x+1} \right| - \frac{4}{3} \left( \frac{1}{x-1} \right) + C \quad \checkmark$$

## Question 2

$$\underline{a} \quad \int_1^2 \frac{dx}{x^2 \sqrt{x-1}}$$

$$= \int_0^{\frac{\pi}{4}} \frac{2 \tan \theta \sec^2 \theta d\theta}{\sec^4 \theta \tan \theta} \quad \checkmark$$

$$= 2 \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta.$$

$$= 2 \int_0^{\frac{\pi}{4}} [1 + \cos 2\theta] d\theta.$$

$$= \frac{\pi}{4} + \frac{1}{2} \quad \checkmark$$

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$$\text{let } x = \sec^2 \theta \\ = 1 + \tan^2 \theta$$

$$\checkmark \quad dx = 2 \tan \theta \sec^2 \theta d\theta$$

$$x = 1 \Rightarrow \theta = 0$$

$$x = 2 \Rightarrow \theta = \frac{\pi}{4}.$$

$$\underline{b} \quad \int_0^1 \sqrt{\frac{3-x}{4+x}} dx$$

$$= \int_0^1 \frac{3-x}{\sqrt{12-x-x^2}} dx \quad \checkmark$$

$$= \frac{1}{2} \int_0^1 \frac{6-2x}{\sqrt{12-x-x^2}} dx \quad \checkmark$$

$$= \frac{1}{2} \int_0^1 \frac{-1-2x}{\sqrt{12-x-x^2}} dx + \frac{1}{2} \int_0^1 \frac{7 dx}{\sqrt{(\frac{7}{2})^2 - (\frac{1}{2}+x)^2}} \quad \checkmark$$

$$= \left[ (12-x-x^2)^{\frac{1}{2}} \right]_0^1 + \frac{7}{2} \left[ \sin^{-1} \frac{2(x+\frac{1}{2})}{7} \right]_0^1$$

$$= \sqrt{10} - \sqrt{12} + \frac{7}{2} \left( \sin^{-1} \frac{3}{7} - \sin^{-1} \frac{1}{7} \right) \quad \checkmark$$

## Question 2

C i) LHS =  $\int_{-a}^0 f(x) dx$  let  $u = -x$   
 $= \int_a^0 f(-u) du$   $\therefore du = -dx$   
when  $x=0 \Rightarrow u=0$   
when  $x=-a \Rightarrow u=a$   
 $\checkmark = - \int_0^a f(-u) du$   
 $= \text{RHS}$

ii)  $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \checkmark$   
 $= \int_0^a f(-x) dx + \int_0^a f(x) dx \checkmark$   
 $= \int_0^a (f(x) + f(-x)) dx.$

iii)  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^5 \cos x) dx = \int_0^{\frac{\pi}{2}} ((x^5 \cos x) + (-x)^5 \cos(-x)) dx \checkmark$   
 $= \int_0^{\frac{\pi}{2}} (x^5 \cos x - x^5 \cos x) dx$   
 $= 0 \checkmark$

### Question 3

a.  $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$

$$= \int_0^{\frac{\pi}{2}} \cos x \cos^{n-1} x \, dx.$$

$$= \int_0^{\frac{\pi}{2}} \cos^{n-1} x \frac{d}{dx} (\sin x) \, dx. \quad \checkmark$$

$$= \left[ \sin x \cos^{n-1} x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x (n-1) \cos^{n-2} x (-\sin x) \, dx \quad \checkmark$$

$$= 0 + (n-1) \int_0^{\frac{\pi}{2}} (1 - \cos^2 x) \cos^{n-2} x \, dx.$$

$$= (n-1) I_{n-2} - (n-1) I_n \quad \checkmark$$

b.  $I_n(1+n-1) = (n-1) I_{n-2} \quad \therefore I_n = \frac{n-1}{n} I_{n-2}.$

Let  $I_6 = \int_0^{\frac{\pi}{2}} \cos^6 x \, dx$

$$I_6 = 5I_4 - 5I_6$$

$$6I_6 = 5I_4$$

$$I_6 = \frac{5}{6} I_4 = \frac{5}{6} \cdot \frac{3}{4} \cdot I_2 \quad \checkmark$$

Now,  $I_2 = \int_0^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 2x \, dx = \frac{\pi}{4} \quad \checkmark$

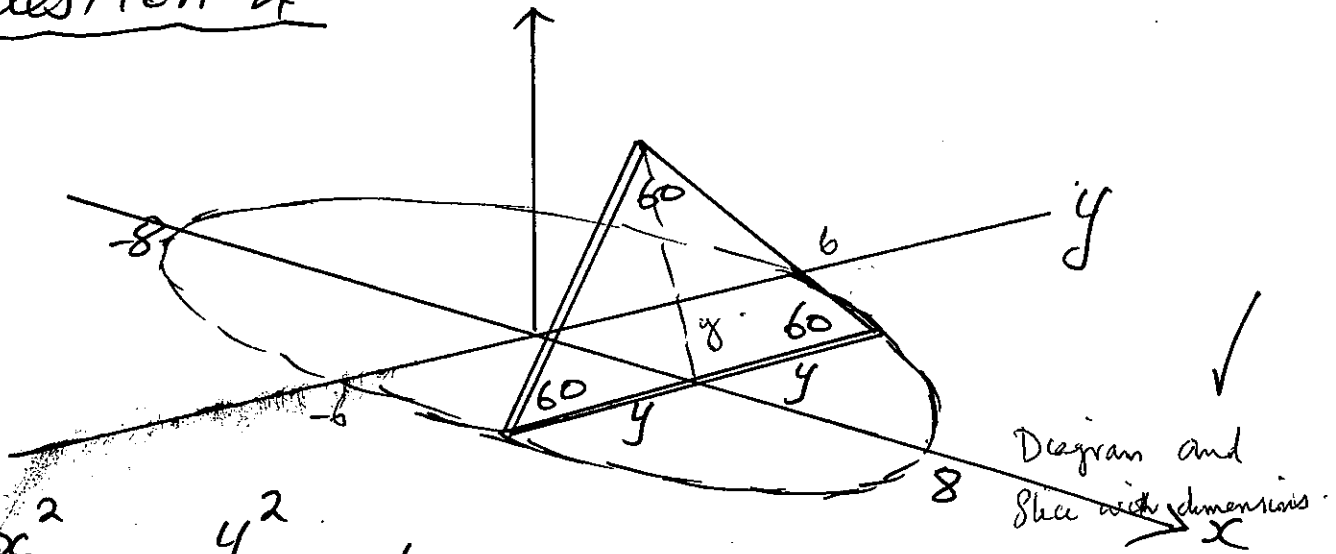
$$I_6 = \frac{5}{6} \times \frac{3}{4} \times \frac{\pi}{4}$$

$$= \frac{15\pi}{96}$$

$$= \frac{5\pi}{32} \quad \checkmark$$

# Question 4

9



$$\frac{x^2}{64} + \frac{y^2}{36} = 1$$

let the surface area of the equilateral triangle,  $A$ , have side length  $2y$ .

$$\text{Area of } A = \frac{1}{2} \times 2y \times 2y \times \sin 60 = \sqrt{3} y^2 \quad \checkmark$$

let Volume of Cross sectional slice be  $dV$

$$\therefore dV = \lim_{dx \rightarrow 0} \sum_{x=-8}^8 \sqrt{3} y^2 dx \quad \checkmark$$

$$\text{Volume} = \sqrt{3} \int_{-8}^8 36 - \frac{9x^2}{16} dx \quad \checkmark$$

$$= 2\sqrt{3} \int_0^8 36 - \frac{9}{16} x^2 dx$$

$$= 2\sqrt{3} \left[ 36x - \frac{9x^3}{16 \times 3} \right]_0^8 = 2\sqrt{3} (288 - 96 - 0)$$

$$= 384 \sqrt{3} \text{ units}^3 \quad \checkmark$$

# Question 4

b.

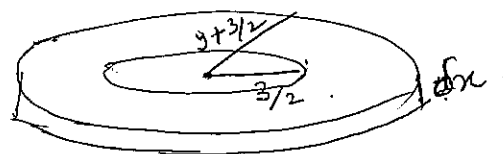
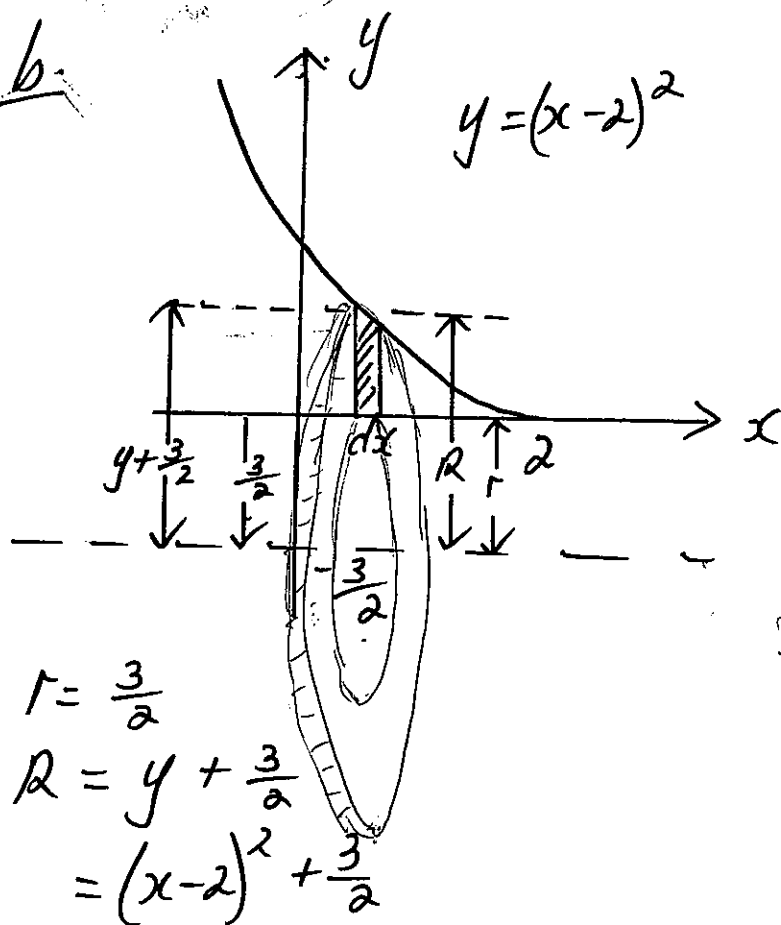


Diagram with slice

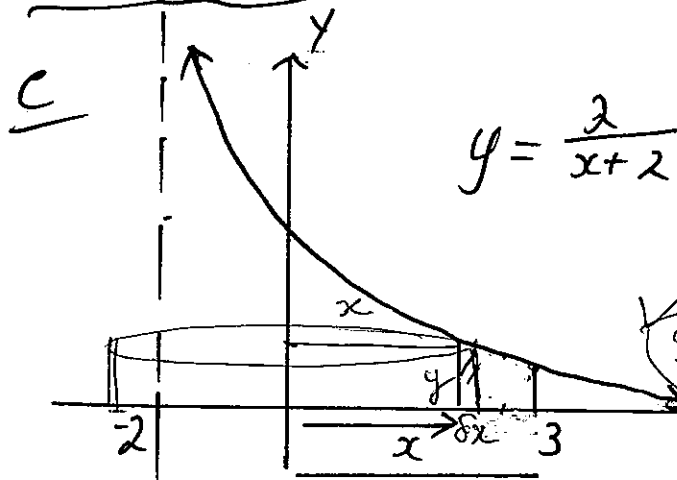
Let  $dv$  be the volume of the annulus with inner radius  $r$  and outer radius  $R$  of thickness  $dx$ .

$$\begin{aligned} \therefore dv &= \pi (R^2 - r^2) dx \\ &= \pi (R - r)(R + r) dx \\ &= \pi y (y + 3) dx \\ &= \pi (x-2)^2 ((x-2)^2 + 3) dx \\ &= \pi [(x-2)^4 + 3(x-2)^2] dx \end{aligned}$$

Volume

$$\begin{aligned} &= \pi \int_0^2 ((x-2)^4 + 3(x-2)^2) dx \\ &= \pi \left[ \frac{(x-2)^5}{5} + (x-2)^3 \right]_0^2 \\ &= 14 \frac{2}{5} \pi \text{ units}^3 \end{aligned}$$

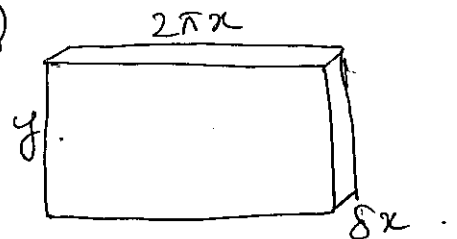
# Question 4



Let the volume of the  
typical shell =  $\delta V$ .

When the shell is opened,  
it forms a rectangular  
prism as shown below.

Diagram  
with shell



$$\therefore dV = 2\pi xy \, dx.$$

$$\text{Volume} = \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 2\pi xy \, dx.$$

$$= \int_0^3 2\pi x \left( \frac{2}{x+2} \right) dx$$

$$= 4\pi \int_0^3 \frac{x}{x+2} dx.$$

$$= 4\pi \int_0^3 \frac{x+2-2}{x+2} dx.$$

$$= 4\pi \int_0^3 \left( 1 - \frac{2}{x+2} \right) dx.$$

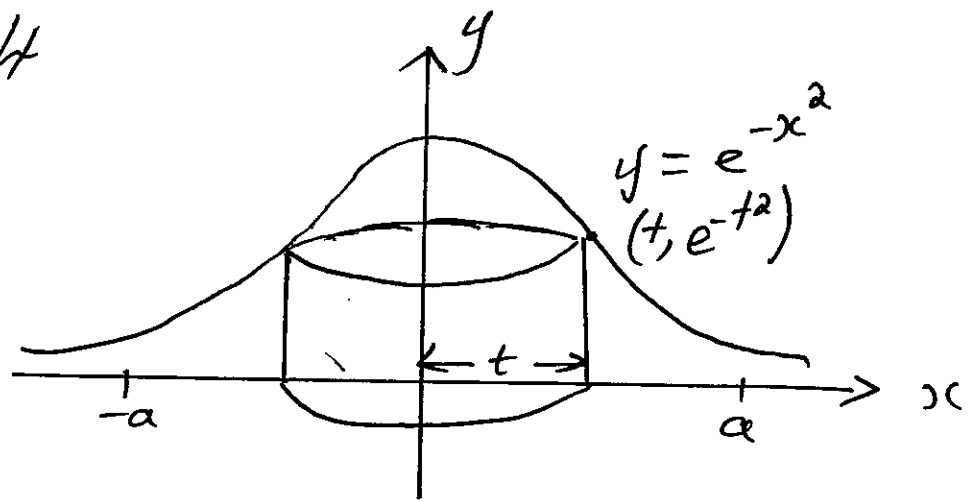
$$= 4\pi \left[ x - 2 \ln(x+2) \right]_0^3$$

$$= 4\pi \left[ 3 + 2 \ln\left(\frac{2}{5}\right) \right] \text{ units}^3$$

# Question 4

d. i

✓



ii let the width of a shell be  $dt$ .

✓

$$\therefore dV = 2\pi xy dt$$
$$= 2\pi t e^{-t^2} dt$$

iii Volume =  $\lim_{dt \rightarrow 0} \sum_{x=0}^a 2\pi t e^{-t^2} dt$  ✓

$$= -\pi \int_0^a -2te^{-t^2} dt$$

$$= -\pi \left[ e^{-t^2} \right]_0^a$$

$$= \pi (1 - e^{-a^2}) \text{ units}^3$$

✓